ML-assisted Segment Routing Traffic Engineering

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1. Description

In real-world network operations, traditional traffic engineering often struggles to react in real time to short-term traffic bursts, which can cause a lag in link adjustments. To address this issue, this project aims to develop a dynamic Segment Routing Traffic Engineering (SR-TE) framework enhanced with machine-learning-based prediction. Instead of relying on simple threshold-triggered mechanisms, the proposed approach trains a predictive model on historical traffic data to learn traffic patterns and forecast link load for the next 5–15 minutes. The system will select the best forwarding path for critical Top-K elephant flows from a set of precomputed candidate paths, enabling traffic reallocation before congestion occurs. By comparing IGP routing and static SR-TE policies in an experimental environment, this project is expected to demonstrate advantages in reducing maximum link utilization and improving tail latency (P99 RTT). In addition, a policy cooldown mechanism will be designed to mitigate network instability caused by frequent path switching, providing practical insights for deploying predictive traffic engineering in production networks.

Keywords – Segment Routing (SR), SR Traffic Engineering (SR-TE), Machine Learning, Elephant Flow (Top-K) Scheduling, Maximum Link Utilization (MLU) Minimization, Tail Latency (P99 RTT) Optimization

2. Literature Review

Segment Routing (SR) is standardized as a source-routing paradigm where the ingress node steers packets through an ordered list of "segments," enabling explicit routing while keeping per-flow state primarily at the ingress [1]. SR can be instantiated over the MPLS data plane (SR-MPLS) [2] and the IPv6 data plane (SRv6) via the Segment Routing Header (SRH) [3]. In practice, SR Traffic Engineering often uses the concept of an SR Policy at a headend node: an ordered list of segments used to steer traffic along a chosen path (often selected from candidate paths) [4]. These foundations make SR-TE a natural actuation layer for automation: once a decision is made, it can be enforced by installing/updating SR policies at ingress nodes [4].

Learning for traffic engineering and prediction-driven control. Recent research has shown that learning can help TE decisions by mapping observed traffic and topology information to routing/allocation actions. For example, TEAL (SIGCOMM 2023) proposes a learning-accelerated TE scheme using a neural approach to allocate demands more effectively under changing

conditions [5]. Meanwhile, forecasting and time-series modeling are well-established: recurrent models like LSTM were designed to capture long/short dependencies in sequences, making them a common baseline for traffic prediction tasks [6]. Feature-based learners such as gradient-boosted trees (e.g., XGBoost) are also widely used due to strong performance and practical training efficiency [7]. A key practical takeaway emphasized in ML-for-networks discussions is that models should be evaluated in end-to-end, operationally realistic loops, not only on offline prediction accuracy [8].

2.1 Our Vision and Solution

Our approach bridges the gap between sophisticated ML theory and stable network operations. The vision focuses on "Selective Orchestration":

Top-K Elephant Flow Scheduling: Instead of recalculating every tiny flow (which wastes CPU cycles), the system identifies and moves only the "Elephant Flows." Controlling the top 10% of flows typically manages 80% of the volume.

Predictive Actuation: We use an XGBoost-based regressor to forecast link utilization. If a link is predicted to exceed a 70% threshold in 10 minutes, the system triggers a path change now.

Stability First: To address the "oscillation" problem found in many ML papers, w e propose a Policy Cooldown Mechanism. This ensures that once a flow is moved, it stays on that path for a minimum duration unless a critical failure occurs, preventing the "thundering herd" effect in route updates.

3. Methodology

3.1 Research Design

We conduct a controlled experimental study comparing three routing/TE approaches under the same topology and traffic: (1) IGP shortest-path routing, (2) static SR-TE, and (3) the proposed ML-assisted dynamic SR-TE. Performance is evaluated quantitatively using standard network KPIs.

3.2 Testbed and Traffic Workload

An SR-capable virtual testbed (e.g., FRR in Mininet/containerlab) is built with multiple viable paths. For each (source, destination) pair, a small set of candidate paths is precomputed. Traffic is generated using time-varying matrices with injected short bursts, including a mix of elephant and mice flows; the controller focuses on Top-K elephant flows per decision period.

3.3 Telemetry Collection

We collect time-series telemetry such as per-link utilization (primary), optional loss/drop counters, and end-to-end RTT (active probing or flow measurements). Data are sampled at short intervals and aggregated into decision windows for model training and policy decisions.

3.4 ML Forecasting

A forecasting model predicts link load (or congestion risk) 5–15 minutes ahead. We start with XGBoost using sliding-window features for robustness and fast iteration, and optionally compare with GRU/LSTM if time allows. Models are trained with time-based splits and evaluated using MAE/RMSE (or F1/AUC for classification). The goal is not to achieve state-of-the-art prediction accuracy, but to evaluate whether simple, robust predictors can improve traffic engineering decisions in an online control loop.

3.5 SR Policy Selection and Stability

At each decision epoch, the system uses forecasts to select SR policies (segment lists) for Top-K elephant flows from candidate paths, aiming to reduce predicted maximum link utilization and improve tail latency.

(For this project, we model the system as a centralized control plane component outside the data plane).

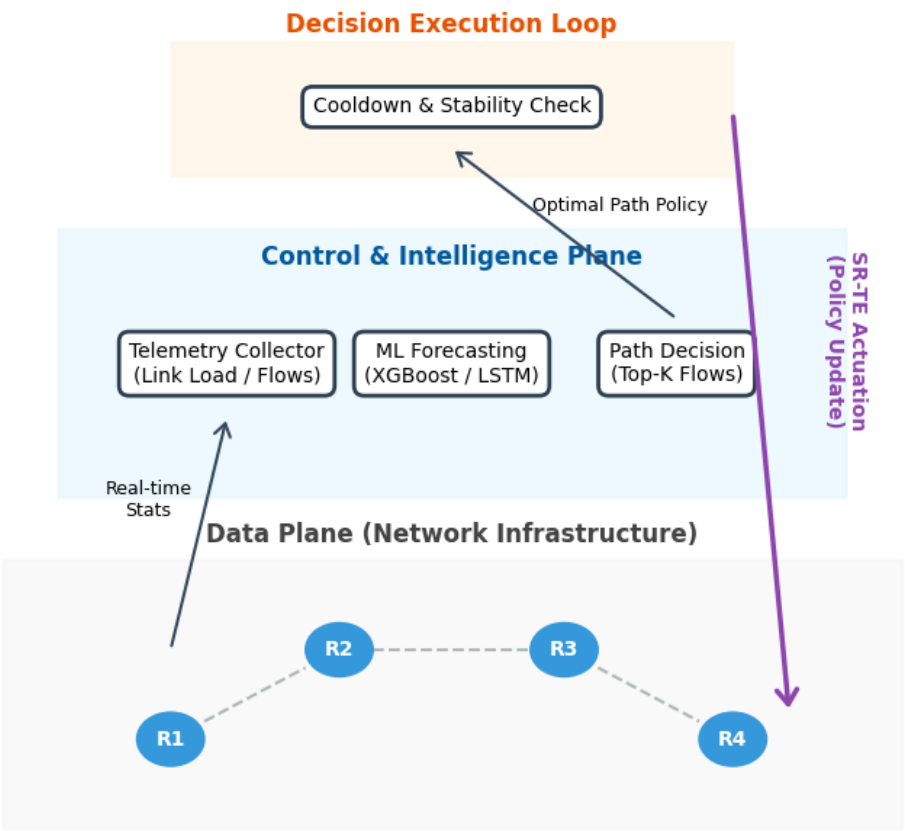
To avoid route oscillation, policy updates are constrained with cooldown and hysteresis (only switch when predicted gains exceed a threshold).

3.6 Evaluation

We compare methods using maximum link utilization (MLU), P95/P99 RTT, loss (if measurable), and operational overhead (policy update rate/oscillation). Additional ablations include varying Top-K and disabling stability controls to quantify their impact.

4. System Block Diagram

To visualize the interaction between the ML controller and the network fabric, the following diagram illustrates the closed-loop architecture:



5. References

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